



# NeuraSign

Physiological intelligence for work

## PROBLEM

# Work is dynamic. So is human capacity.

## WHAT COMPANIES KNOW

Skills



Role



Calendar



Deadlines



## THE MISSING CONTEXT

Fatigue



Focus



Stress



Cognitive load



Readiness



The risk: **lost time** and **costly errors**.

# Amazon

# \$56M

Jury award

Loss of alertness.  
Head-on collision.

Amazon delivery partner driver

Crash 16 Nov 2024

Verdict Jul 2026

INSURANCE JOURNAL

## Jury Awards 78-Year-Old Victim \$56 Million for Crash Caused by Amazon Delivery Driver

By Andrew G. Simpson | July 27, 2026



Article photo: Amazon

# Dry Tortugas

# \$5M

Vessel and platform damage

Extra workload.  
Asleep at the helm.

Reported sleep: 8–9 hours a day

Collision 16 Aug 2025

NTSB report 26 Aug 2026

NATIONAL TRANSPORTATION SAFETY BOARD



Left to right: The *Dry Tortugas* underway in January 2024, and platform VK-817A at an unknown date after the casualty. (Sources: Scott Rambo, marinetraffic.com, and Third Coast)

## Contact of Offshore Supply Vessel Dry Tortugas with Offshore Platform VK-817A

# Same skills. Different capacity.

Same person, different moments



 Lower cognitive load

 Higher cognitive load



Physiological signals

+ personal baseline

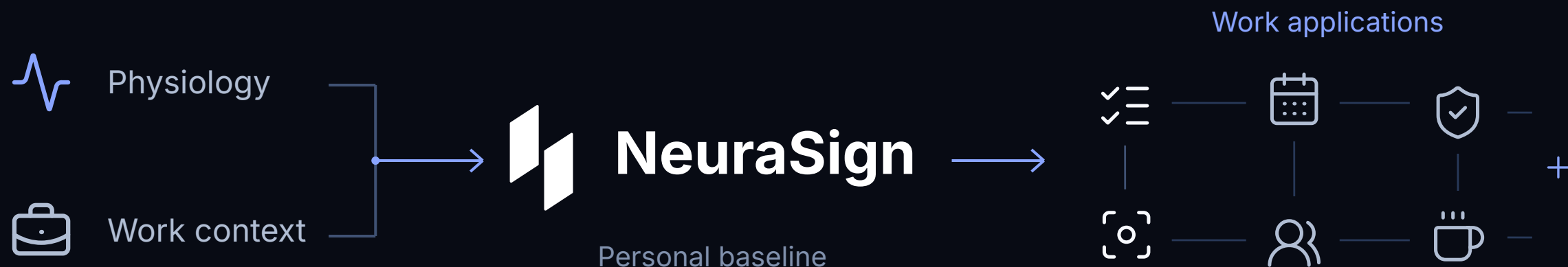


**State estimates**



**Work decisions**

# Human-state intelligence for adaptive work.



TARGET PRIORITY

01  **High-risk operations**

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02  **High-pressure environments**

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03  **High-performance teams**

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## Scientific evidence

### Evidence for state estimates

 **Stress** HRV + EDA  
Schmidt et al. (2018)

 **Mental fatigue** HR / HRV  
Matuz et al. (2021)

 **Workload** HRV + EDA  
Hirachan et al. (2022)

 **Readiness\*** HRV  
Kiviniemi et al. (2007)

\* Indirect support from  
physical-training research.

## Market adoption



**~300**

EGA workers  
Physiological wearables

Industrial heat safety  
Summer 2024

Emirates Global Aluminium

## Our proof

Lab stress / WESAD

**94.2%**

Selected-model accuracy  
80% train / 20% test (people)  
96.5% logistic reference

Daily Oura score

**~4** points / 100

Reported prediction error

Offline research  
Dashboard integration pending

01

## Dynamic task assignment

SOC + hospital emergencies

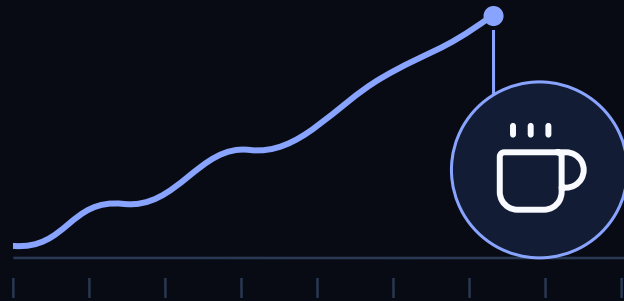


Suggest the next responder.

02

## Overload prevention

Long hospital shifts



Prompt a break or rotation.

03

## Safety-critical operations

Surgery



Support a pre-op team check-in.

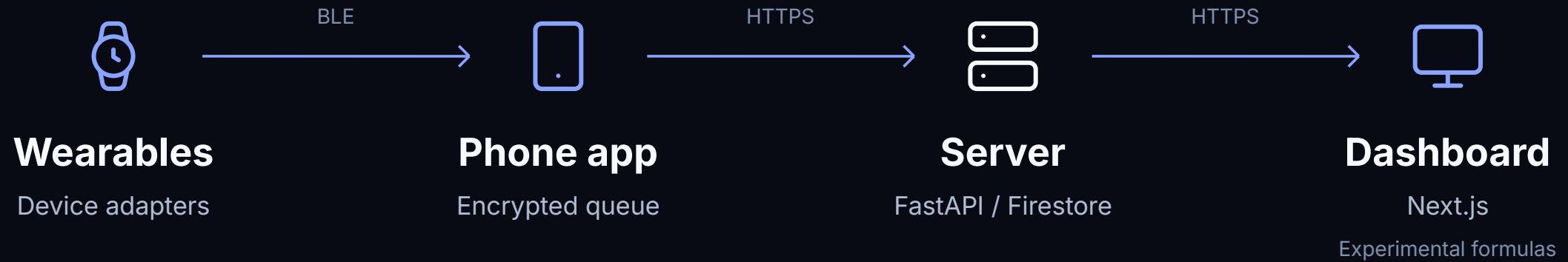
COMPETITION & DIFFERENTIATION



Qualitative product positioning. NeuraSign shows product vision.

## TECHNICAL ARCHITECTURE

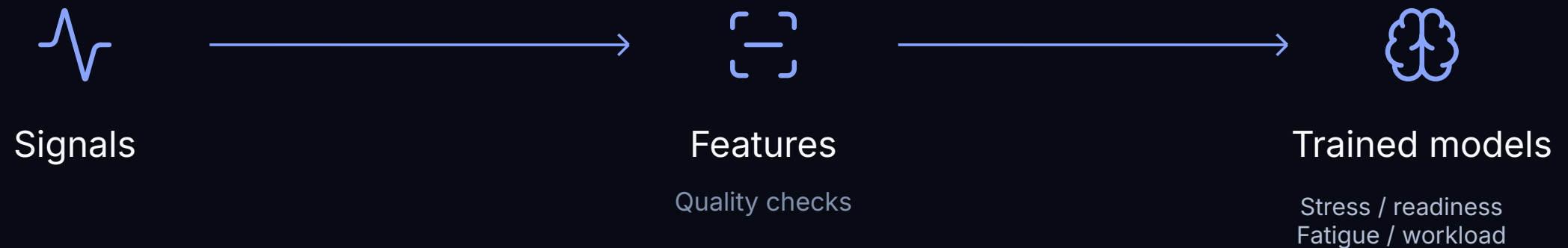
### CONNECTED PLATFORM



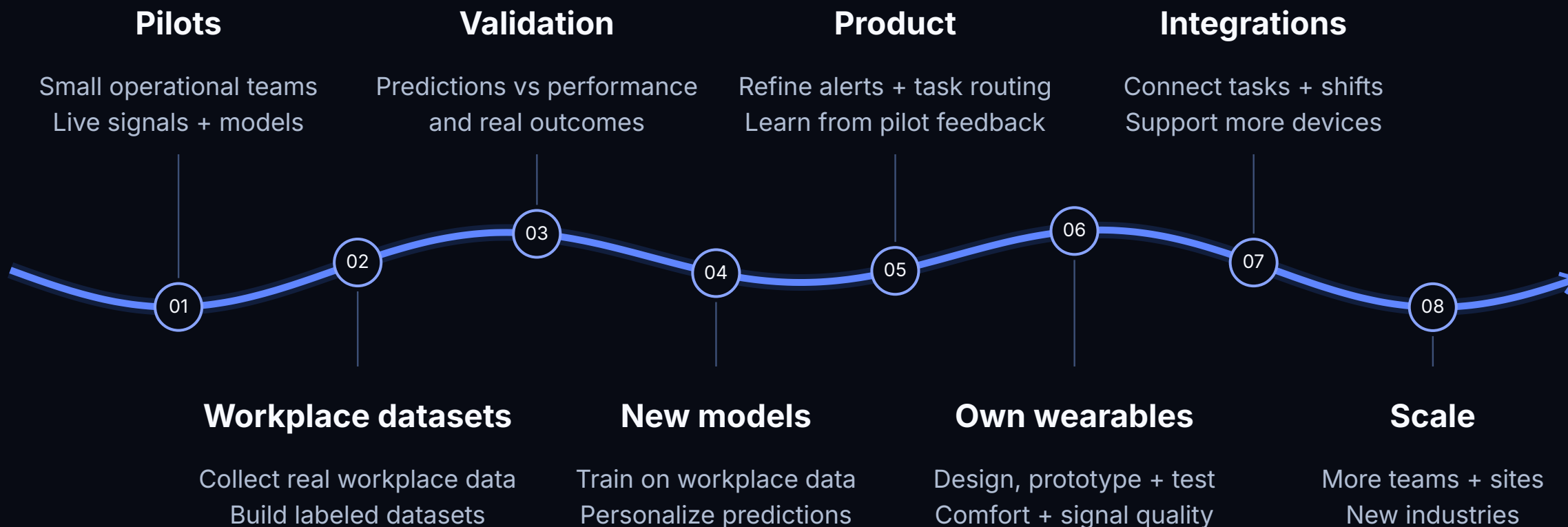
### Research engine

Python / scikit-learn / CatBoost

Model integration next



# Work that adapts to human capacity.



# Annual software. Per active participant.

State insights, work recommendations and workflow integrations



## Paid pilot

Defined scope + success criteria  
Pilot fee credited on conversion

## Annual site contract

Per participant + site minimum  
Recurring software subscription

## Multi-site expansion

More participants + locations  
Reusable workflow integrations

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### TARGET BUYER

Operations + safety leaders

### ADDITIONAL SERVICES

Onboarding + custom integrations

### HARDWARE

Compatible devices first

Proposed model. Pricing and contract terms to validate in pilots.

# Control over data and decisions.

## IMPLEMENTED IN THE PROTOTYPE

### **Pause & disconnect**

Stop phone capture and uploads

### **Scoped access**

Company + team permissions

### **Protected transfer**

Encrypted queue + HTTPS

Authorized managers can view individual measurements, including raw samples.

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## PROPOSED PILOT COMMITMENTS

### **Voluntary participation**

No penalty for declining  
or withdrawing

### **Purpose limits**

Minimum necessary data  
No ranking or disciplinary use

### **Accountable decisions**

Human review and challenge  
No fitness-for-work judgments

Before pilots: interpretation-only manager access and a deletion route for every participant.

# Model evaluation

Offline tests with people separated across splits

Stress / WESAD

## 94.2%

accuracy / rest vs lab stress

96.5% logistic reference

3 held-out people / 60-second windows

Readiness / daily Oura score

## ~4

points error / 100

Approximation of a daily vendor score

Instant readiness remains unvalidated

Fatigue

## 63.7%

balanced accuracy

66.7% previous model / same evaluation

Reliability remains insufficient

Workload / NASA-TLX

## ~13

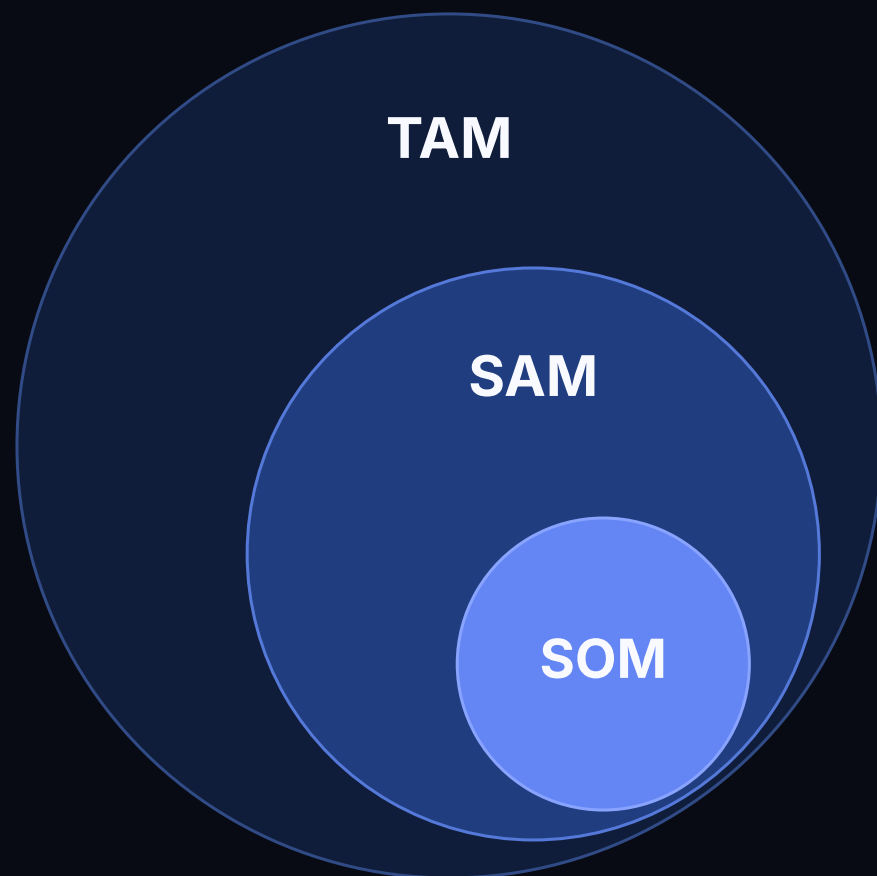
points error / 100

Similar error using only rest vs activity

Added value from physiology unproven

# U.S. TAM / SAM / SOM

Illustrative pricing: \$20 per participant / month



BLS QCEW 2025 / Circles not to scale

## TAM / SECTOR CEILING

**\$4.9B** / year

20.37M U.S. jobs × \$240/year

Manufacturing, transport, mining + utilities

## SAM / ELIGIBILITY SCENARIO

**\$244M** / year

1.02M eligible seats × \$240/year

5% of sector jobs assumed eligible

## SOM / INITIAL ROLLOUT

**\$240k** / year

10 sites × 100 participants × \$240/year

Illustrative 24-month deployment scenario

SAM eligibility and SOM rollout are planning assumptions, not validated demand or a sales forecast.

# Where the products overlap.

| Product                                                                                                                                | Scope              | Primary inputs                        | Decision support                     |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------|--------------------------------------|
|  F <i>FATIGUE</i><br>SCIENCE                          | Workforce fatigue  | Sleep + work schedules                | Forecasts, breaks + reassignment     |
|  SMART<br>CAP<br><small>Home Safe - Every Day</small> | Operator alertness | EEG + operational data                | Alerts, escalation + task rotation   |
|  SOTER                                                | Workplace safety   | Motion + task video                   | Coaching + corrective actions        |
|  CATAPULT                                             | Sports teams       | Location, motion + HR                 | Training load + return to play       |
|  OURA                                                 | Organizations      | Ring physiology + sleep               | Group readiness + resourcing         |
|  WHOOP®                                             | Consumer           | Physiology + training                 | Daily effort + recovery coaching     |
|  NeuraSign                                          | Product vision     | Physiology + skills<br>+ task context | Work allocation + adaptive workflows |

Vendor capabilities shown as documented. NeuraSign row describes the proposed product.

Sources: official product documentation, Sep 2026. Comparative effectiveness has not been established.

# Supervised learning



## WESAD / held-out test

**12** development



**3** held out



**434**

windows

**60s**

per window

**57**

features

32 configurations / 4 person-grouped folds

Selection uses development participants only.

## Latest models / nested validation

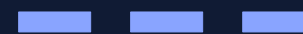
Readiness / Fatigue / Workload

**4×**

outer folds

Each outer fold

3 inner folds



Held-out  
people

Inner folds select. Outer folds evaluate.

People remain separate in every split.